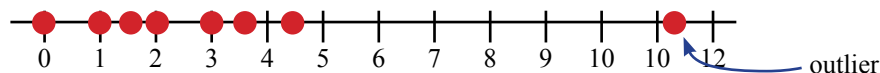




Outliers



An outlier (or rogue value) is a piece or pieces of data that are *extremely different* from the vast majority of the data set.



A. Spot the outlier in each set of data.

- 1). 25, 12, 19, 23, 11, 26, 17, 191, 30.
- 2). 320, 415, 297, 13, 368, 305, 395, 408, 356.
- 3). 883, 1002, 835, 291, 946, 813, 983.
- 4). 1342, 1291, 1513, 1296, 4001, 1493, 1462.
- 5). 0.003, 0.007, 0.005, 0.4, 0.006, 0.004.
- 6). 3005, 5099, 3873, 5354, 4012, 127001, 4978.

B. Spot the outlier in each set of data.

Copy and complete the table for questions 1).-6)..

- 1). 7, 5, 9, 9, 12, 26, 9, 7, 6.
- 2). 83, 81, 69, 79, 12, 94, 79, 79, 81.
- 3). 17, 11, 18, 18, 12, 36, 18, 17, 15.
- 4). 170, 129, 120, 189, 120, 16, 150, 139, 137.
- 5). 31, 38, 41, 49, 180, 32, 35, 37, 43.
- 6). 73, 14, 78, 59, 52, 55, 69, 67, 55.
- 7). Given the following data: 125, 149, 128, 49, 149, 163, 116, 129.

	with outlier	without outlier
mean		
median		
mode		
range		

- a). What effect will the outlier have on the median of the data if the outlier is excluded?
- b). What effect will the outlier have on the mean of the data if the outlier is excluded?
- c). What effect will the outlier have on the mode of the data if the outlier is excluded?
- d). What effect will the outlier have on the range of the data if the outlier is excluded?

8). What will be the variation in the mean temperature if the outlier is included?

110°F, 130°F, 73°F, 129°F, 128°F, 118°F, 117°F.

9). Describe the effect of an outlier on the mean, median, mode and range.

10). If there is an outlier in the data, which measure of central tendency (mean or median) should you use to represent the data?

C.

- 1). a). What is the mean population of all the cities in the table?
- b). If the population of lowest outlier is not included, what is the mean population of the other 5 cities?
- c). If the population of highest outlier is not included, what is the mean population of the other 5 cities?
- d). What is the mean population of cities excluding the highest and lowest outliers?

Cities	Population (millions)
Antwent	5.24
Boomtown	9.58
Classton	0.57
Dippeldorf	3.94
Entchester	4.05
Frankbury	5.84

2). In a maths test 20 students scored the following:

47, 39, 40, 45, 43, 42, 46, 46, 43, 39, 41, 46, 43, 47, 44, 38, 40, 35, 6, 50.

By how much is the mean increased if the outlier is not included?

3). Mr James excluded 3 students who scored low in his science class and said the mean of 15 students was 75 marks. The actual mean of marks of 18 students was 65 marks.

What is the mean of the 3 low scores?

4). The maths scores of 10 students in class are 38, 39, 45, 30, 36, 93, 45, 46, 31 and 32.

By how much is the mean decreased if the outlier is not included?

5). 25 students scored a mean of 45 marks on a science test. 5 students scored extremely high marks. When these 5 students were excluded the mean was reduced by 5 marks.

What was the mean score of the 5 high scoring students?

6). 40 children guessed the number of marbles in a jar and the average guess was 326.

However three of the guesses were way too high and so were excluded from the competition.

When these three outliers were excluded, the mean guess was reduced by 72.

What was the mean of the three outliers?